



UNIVERSITY OF CALGARY

**“Automated Congestion and Curtailment Screening for
Renewable Transmission Corridors in Southeast Alberta”**

Final Project Report

**Graduate Project in Electrical Engineering
ENEL 678**

Group: 06

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1. Glossary

Term	Definition
AIES	Alberta Interconnected Electric System.
AESO	Alberta Electric System Operator, the source of the hourly load, generation, and planning criteria used in this project.
ACCOR	Automatic Congestion Relief function in PSS®E used to automate generator curtailment for overload relief.
N-0	Normal operating condition with all modelled network elements in service.
N-1	Single-element contingency condition in which one transmission element is removed from service.
AC power flow	Steady-state alternating-current load-flow analysis that captures both active-power and reactive-power behaviour, including bus voltages.
SE Alberta	Southeast Alberta transmission corridor, the project's main reporting focus.
Proxy dispatch	A substitute hourly generation profile assigned to a modelled generator that lacks its own historical hourly profile.
Constant-PQ scaling	Load scaling approach in which both MW and Mvar are scaled proportionally so the base-case power factor is preserved.

2. Executive Summary

This project develops a planning-level, Python–PSS®E–based screening tool for evaluating transmission congestion, voltage performance, and resultant curtailment amount in generators in renewable-dominated Alberta Transmission systems. The central engineering problem is that manual or case-by-case studies are not efficient for evaluating a full year of system conditions when generator generation and load both vary hourly. A more scalable and reproducible workflow is required to identify where congestion happens, how often it occurs, and how much generation curtailment is needed to relieve it and if the curtailment can relieve all overloads.

The proposed solution combines three layers:

- I. Data preparation,
- II. Automated power-flow execution, and

- III. post-processing /reporting which allows user to see how much generation curtailed in generator hourly.

At the very first, historical AESO hourly load and generation data [1,2] are mapped to a solved Alberta PSS®E base case [3]. The tool then applies 8760 hourly operating points, runs N-0 AC power flow, screens thermal and voltage violations, invokes ACCOR where overload relief is needed, and stores structured outputs for later ranking and visualization.

The quantitative evidence in this report is driven by the finalized N-0 summary workbook (see Supplementary Files). The solved N-0 dataset contains 5215 hourly cases. Within that solved set, the highest observed pre-ACCOR loading is 165.01%, the average hourly maximum loading is 136.46%, the maximum simultaneous thermal-violation count is 9, and the maximum simultaneous voltage-violation count is 587. The dominant recurring thermal bottleneck is branch 816-817, which is overloaded in 5128 solved hours, while the voltage profile is dominated by persistent high-voltage excursions at a limited set of buses, most notably bus 10605 (THICKWD).

3. Project Motivation and Objectives

The Alberta electricity system has experienced rapid growth in wind and solar integration, particularly in the Southeast Alberta corridor. That transition is beneficial from a de-carbonization perspective, but it changes power-flow patterns in ways that can increase transmission stress during high-renewable and low-demand operating conditions. Under those conditions, planners may observe recurring thermal overloads, voltage deviations, and economically significant renewable curtailment.

Traditional planning workflows often rely on a limited number of snapshots or manually assembled studies. Those methods are useful for targeted engineering checks, but they are not well suited to revealing recurring congestion patterns over a full year. They may miss hours in which the system is not at annual peak yet remains highly stressed because of renewable concentration, corridor topology.

The primary objective of this project is therefore to create an automated tool that executes hourly AC power-flow analysis for a full Alberta transmission model, detects thermal and voltage violations, applies ACCOR-based curtailment where needed, and summarizes the resulting congestion and curtailment patterns in a form suitable for early-stage planning and corridor assessment.

- Execute 8760 hourly N-0 AC power-flow simulations using a solved Alberta PSS®E case and mapped AESO hourly inputs.

- Identify overloads on monitored transmission lines at 69 kV and above and detect bus-voltage violations relative to planning limits.
- Quantify generator curtailment by hour and by generator after ACCOR-based overload relief, with filtering capability for user-selected generators.
- Rank congestion-limiting transmission elements by both frequency and severity so that the most persistent bottlenecks become visible.
- Generate structured data files, summary tables, and report-ready figures that support planning interpretation rather than real-time control.

4. Methodology and Design

The final methodology is intentionally modular so that the tool can be replicated, audited, and extended. The design is divided into three layers:

- I. input and data preparation,
- II. automation and screening, and
- III. post-processing and reporting.

This separation reduces debugging complexity, improves traceability, and makes it easier to insert additional functionality later without restructuring the workflow.

4.1 Overall System Architecture

The architecture begins with a solved Alberta PSS®E base case and hourly AESO load and generation datasets. These inputs are transformed into case-consistent hourly dispatch and load tables through mapping and validation (see appendix A, B). The automation layer then applies those hourly conditions to the base case, solves the network, detects violations, invokes ACCOR when needed, and records both raw and aggregated outputs. The reporting layer converts those outputs into ranking tables, curtailment summaries, and voltage-screening results.

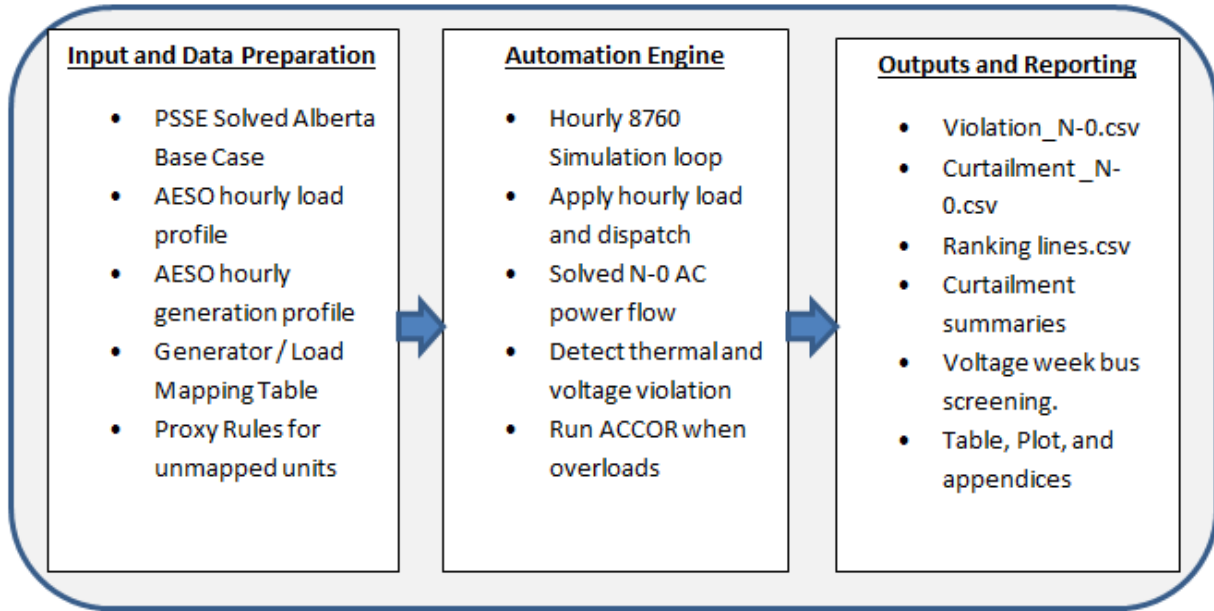


Figure 1: Conceptual architecture of the automated congestion screening tool.

4.2 Inputs and Data Sources

The required inputs are:

- (i) a solved Alberta transmission base case in PSS®E,
- (ii) historical AESO hourly load data,
- (iii) historical AESO hourly generator metered volumes,
- (iv) mapping tables between AESO identifiers and PSS®E buses or machine IDs, and

No dynamic models are required because the project is scoped as a steady-state planning study.

The solved base case acts as the anchor condition for every hourly simulation. Instead of rebuilding network data for each hour, the tool reuses the same case topology and device settings and updates only the operating point. This preserves consistency across the year and aligns the study with its planning-screening purpose.

4.3 Load Modelling Methodology

Load modelling follows an area-based scaling approach built on the solved base case. Base-case constant-power loads are aggregated by PSS®E area to obtain reference MW and Mvar totals. AESO historical hourly area demand is converted into normalized hourly shapes so that temporal variation is

captured separately from absolute demand level. A study-year growth factor [4] is then applied to shift the base-case load level to the intended study year.

The resulting hourly scaling factors are applied to each load record within the corresponding area while preserving base-case spatial allocation and power factor. In practical terms, each bus-level load is treated as a fixed share of its area's total demand, and both MW and Mvar are scaled by the same hourly factor. This constant-PQ assumption is appropriate for a planning-level screening tool because it preserves the reactive-to-real-power relationship embedded in the solved case. (Details in Appendix A)

4.4 Generation Dispatch Modelling Methodology

Generator dispatch modelling is anchored to the same principle of case consistency. Each hourly AESO asset is mapped, where possible, to one or more machines in the Alberta PSS®E base case. If a single hourly asset corresponds to one modelled machine, its hourly MW value is applied directly, subject to machine capability and in-service status. If a single hourly asset corresponds to multiple modelled machines at a common point of interconnection, the dispatch is split across those machines in proportion to capability.

Two additional rules make the method robust. First, hourly assets that are not represented in the base case are excluded rather than injected artificially, which avoids hidden balancing through slack behavior and preserves transparency. Second, generators present in the base case but lacking hourly AESO profiles receive proxy dispatch shapes based on similar technology and the same or nearest planning area.

4.5 Screening Criteria and Study Assumptions

Monitored elements are transmission lines rated at 69 kV and above; transformers are intentionally excluded from the baseline implementation so the analysis remains focused on transmission-line congestion. Thermal violations are flagged when line loading exceeds 100% of the applicable continuous rating. Emergency ratings are not used in the baseline planning assessment.

Voltage violations are screened using normal planning limits for N-0 conditions. For normal operation, the baseline uses a voltage band of 0.95 to 1.05 p.u.

4.6 Automation Architecture and Code Modules

For report clarity, the two operational Python scripts were segmented into appendix-ready modules that mirror the planned code architecture. This packaging does not change the underlying logic; instead, it makes the final workflow easier to review, audit, and maintain.

Module	Primary responsibility
config.py	Stores project paths, study thresholds, ACCOR option arrays, and control labels.
io_aeso.py	Loads hourly AESO-style CSV inputs, standardizes timestamps, and prepares dispatch tables.
mapping.py	Formats machine IDs and builds AESO-to-PSS®E generator-name lookup tables.
psse_api.py	Contains wrapped PSS®E calls for initialization, output control, solving, and machine snapshots.
screening.py	Extracts branch and bus metrics and identifies overload and voltage violations.
accor_control.py	Builds DFAX inputs, executes ACCOR, and writes debug and text-log files.
contingency.py	Reserved modular placeholder for selected N-1 logic in the appendix package.
main.py	Orchestrates the hourly loop, error handling, metric collection, and result writing.

Table 1: Automation Architecture and Code Modules

4.7 Hourly N-0 Workflow

For each hour of the year, the tool starts from the solved base case, applies the hourly load profile, applies the hourly generator dispatch, and runs AC power flow. If the case does not converge, the hour is logged, flagged for review, and the tool continues to the next hour instead of terminating the full annual run. If the case converges, the tool checks the monitored transmission elements and bus voltages. When overloads are present, ACCOR is invoked to produce a curtailment-based relief action while maintaining solvability.

This continue-on-error design is one of the strongest design choices in the project because it prevents a limited number of difficult hours or bad inputs from invalidating the entire study. It also creates a transparent record of where the system or data requires further engineering review.

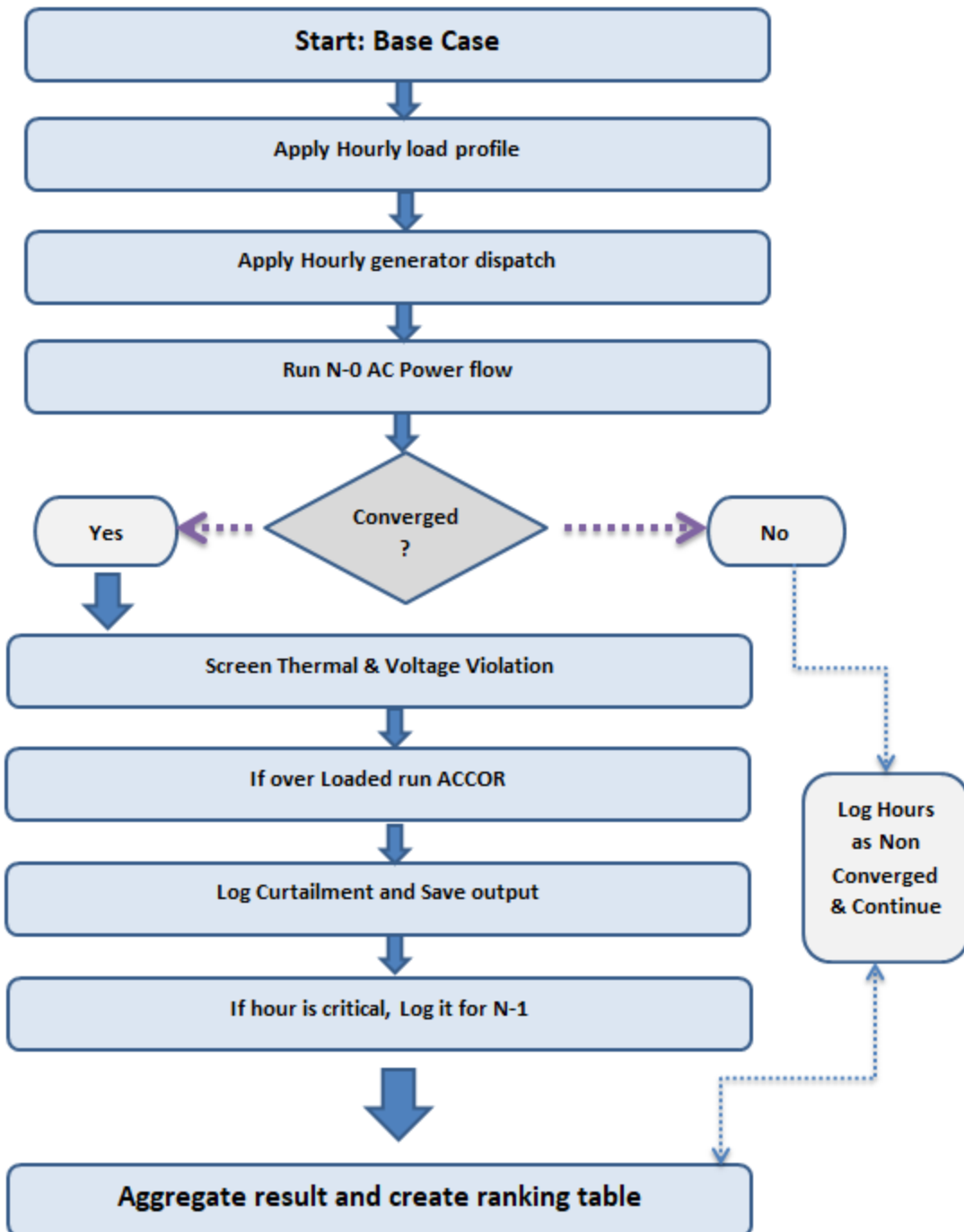


Figure 2: Hourly simulation and screening workflow for the baseline N-0 study.

4.8 Post-Processing, Ranking, and Reporting

The post-processing layer converts raw simulation outputs into planning insights. At minimum, the tool stores N-0 violation logs, post-ACCOR status logs, curtailment records, and line-ranking tables. From those structured outputs, the report derives top congested lines by frequency, lines by maximum overload, persistent voltage-violation buses.

The updated N-0 summary workbook (see Supplementary files) used in this final report was intentionally pruned to keep only the most report-relevant sheets.

4.9 Main Gist of the Python Implementation

- The code begins by initializing PSS®E, muting console output, and defining paths, thresholds, and ACCOR control settings.
- Hourly load and generation input files are read using pandas, timestamps are normalized to hourly resolution, and generator identifiers are standardized for mapping to PSS®E machines.
- For each hour, the base case is reloaded, hourly load is applied with `load_chng_5`, and generator MW dispatch is applied with `machine_chng_2`.
- The solver uses a fallback strategy: flat-start Newton-Raphson, normal Newton-Raphson, `mslv`, and a final retry. This improves robustness over long batch execution.
- After each successful solve, branch flows and bus voltages are extracted using `abnreal`, `abrncar`, `abrnint`, `abusreal`, and `abusint`. The code then computes maximum loading, thermal violation records, and voltage-violation records.
- If thermal violations exist, DFAX data are built and ACCOR is executed. Detailed ACCOR report, progress, alerts, prompt, and debug-summary text files are written for traceability. (see supplementary files)
- Pre- and post-ACCOR generator snapshots are compared to estimate curtailment at machine level. Finally, all major outputs are exported to CSV for downstream ranking and report preparation.

5. Product Scope and Final Product Functionality

The final product is a software tool for planning-level congestion and curtailment screening. It is not a market simulator, not a transient-stability tool, and not a real-time operational dashboard. Its purpose is to help planners and researchers identify recurring transmission stress and generator curtailment exposure over a large annual operating horizon using consistent and repeatable assumptions.

Relative to the original roadmap, the most important scope clarification is that the project’s engineering value is concentrated in the automated N-0 pipeline, ACCOR-based curtailment. Features such as an interactive dashboard are also included.

Functionality	Status	Comment
8760-hour N-0 AC power-flow screening	Core	Fundamental baseline capability of the tool.
Transmission-line congestion monitoring (≥ 69 kV)	Core	Primary monitored element set; aligned with corridor focus.
Bus-voltage screening	Core	Required to complement thermal screening and preserve AC planning relevance.
ACCOR-based congestion relief and curtailment logging	Core	Implemented in code; quantitative reporting is currently strongest for the N-0 pre-ACCOR baseline.
Interactive dashboard / result explorer	Completed	To provide a user interface to see curtailment amount by generators.

Table 2: Features such as an interactive dashboard

6. Technical Specifications of the Final Product

6.1 Power-System Modelling Environment

- Simulation platform: PSS®E steady-state AC power flow.
- Network model: full Alberta Interconnected Electric System transmission base case.
- Operating states: N-0 for all hours;
- Temporal coverage target: 8760 hourly operating conditions representing a non-leap-year 2025 study horizon.

6.2 Data and Mapping Specifications

- Input dispatch basis: AESO hourly generator metered volumes, mapped to modelled PSS®E machines.
- Input load basis: AESO hourly area demand profiles applied through constant-PQ scaling of base-case loads.

- Generator mapping rules: direct match, multi-machine split, exclusion of non-modelled assets, and proxy assignment for modelled assets without hourly profiles.
- Traceability requirement: non-convergent hours and ACCOR actions are logged explicitly.

6.3 Screening and Curtailment Specifications

- Monitored elements: transmission lines rated 69 kV and above.
- Thermal criterion: overload flagged when line loading exceeds 100% of continuous rating.
- Normal voltage criterion: voltage violation flagged when $V < 0.95$ p.u. or $V > 1.05$ p.u. under N-0.
- Curtailment engine: PSS®E ACCOR applied system-wide when overload relief is required.

6.4 Output Specifications

- Structured output files include hourly summary CSVs, thermal and voltage violation CSVs, curtailment logs, ACCOR debug logs, and ranking tables.
- Primary summary products are top congested lines by frequency and severity, persistent voltage-violation buses.

6.5 Deviations from the Design Roadmap

- The roadmap envisaged a broader validated N-1 result package and optional interactive exploration tools. In the current final report, the validated quantitative evidence is strongest for the N-0 baseline. Selected N-1 methodology and hour selection are preserved, but the completed N-1 numerical result set is not completed.
- Transformer monitoring remains excluded from the baseline specification. That is a deliberate design decision rather than a shortcoming, because the project is focused on corridor-level transmission-line congestion and generator curtailment.

7. Measuring Success and Validation Test Results

This section shows finalized N-0 evidence available from N-0 summary workbook (SEE SUPPLEMENTARY FILES). The quantitative discussion below is intentionally focused on the validated N-0 baseline.

7.1 Final N-0 Results Summary

Metric	Value	Unit
Solved / analyzed N-0 hours	5215	hours
Maximum pre-ACCOR loading	165.01	%
Average of hourly maximum loading	136.46	%
Maximum simultaneous thermal violations	9	count
Average simultaneous thermal violations	4.30	count
Maximum simultaneous voltage violations	587	count
Average simultaneous voltage violations	537.99	count
Number of violated branches	15	branches
Number of violating buses	744	buses

Table 3: Final N-0 Results

The current validated N-0 result set contains 5215 solved hourly cases. This is below the original 8760-hour target, which means the report must be interpreted as a partial but still engineering-relevant annual baseline. Within that solved set, the network exhibits materially elevated stress: the average hourly maximum loading is 136.46%, and the maximum observed pre-ACCOR loading reaches 165.01%.

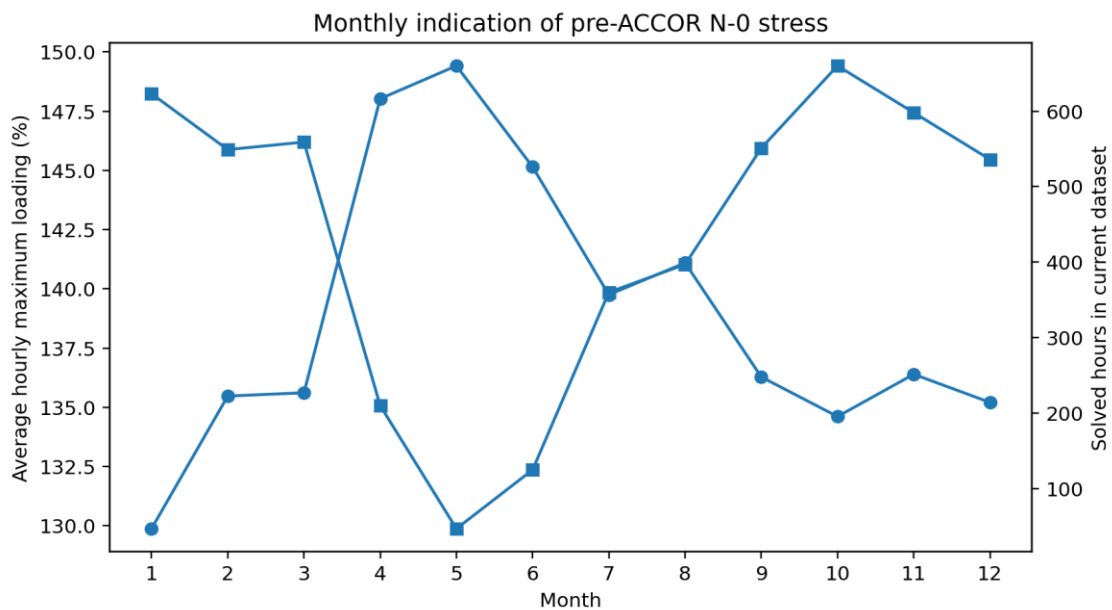


Figure 3: Monthly indication of pre-ACCOR N-0 stress

Monthly coverage is uneven in the current solved dataset, so temporal interpretation must be cautious. Even so, the figure shows that severe stress is not isolated to a single season. Elevated average maximum loading appears across multiple periods, with especially strong values in April, May, and October within the available solved-hour set.

7.2 N-0 Thermal Results

Thermal ranking was performed using pre-ACCOR N-0 violation records. Persistence is measured by the number of solved hours in which a branch is overloaded, while severity is reflected by peak loading and cumulative excess above 100% loading. This gives a more complete view than a single worst-case number because it distinguishes recurring bottlenecks from rare but sharp spikes.

Branch	Circuits	Violated hours	Share of solved hours (%)	Maximum loading (%)	Average loading (%)
816-817	87	5128	98.33	165.01	136.32
817-818	87	4710	90.32	150.72	123.07
12-412	20	4117	78.95	128.89	114.00
818-819	87	3617	69.36	139.97	114.69
816-1501	87	1436	27.54	111.85	103.69
31-272	10	1323	25.37	142.60	117.56
20-88	40	921	17.66	107.07	102.38
329-819	87	789	15.13	133.55	104.51
873-875	79	240	4.60	102.15	101.83
20-39	02	72	1.38	103.07	101.23

Table 4: N-0 Thermal Results

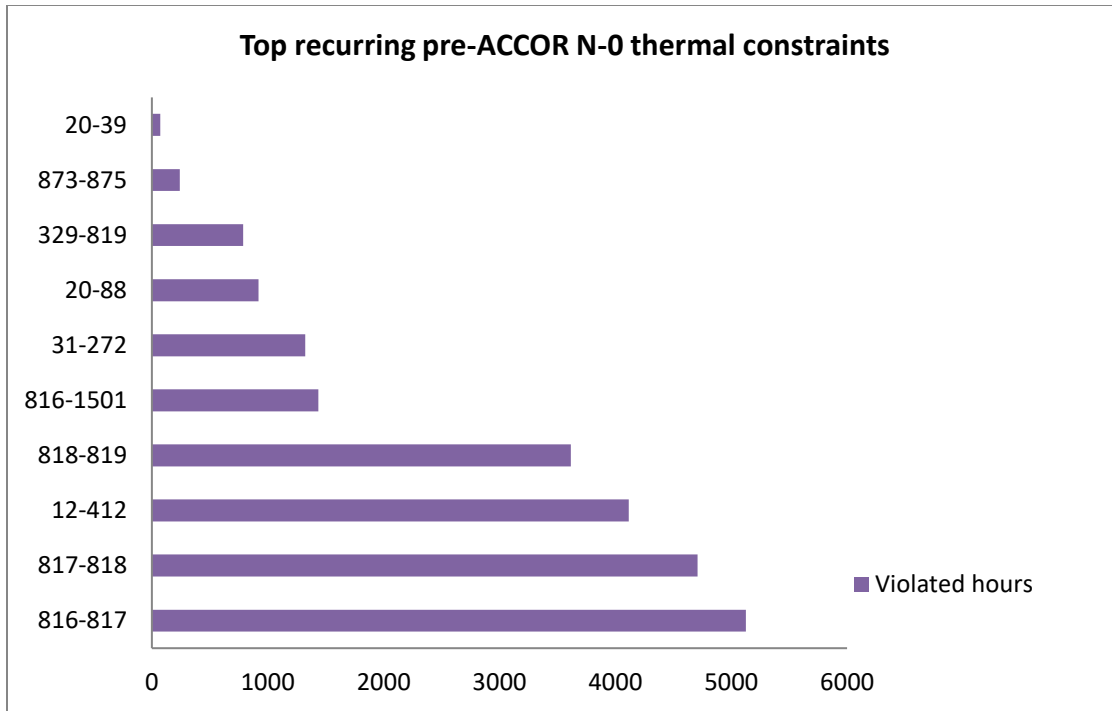


Figure 4: Top recurring pre-ACCOR N-0 thermal constraints by violated hours

The dominant bottleneck is branch 816-817. It is overloaded in 5128 of the 5215 solved hours, corresponding to 98.33% of the solved dataset, with a peak loading of 165.01% and an average overloaded-state loading of 136.32%. The second-ranked branch, 817-818, is overloaded in 4710 solved hours and reaches 150.72%. These two lines, together with 12-412 and 818-819, clearly define the core recurring thermal-stress corridor in the available N-0 results.

A second engineering observation is that some branches are severe because of persistence while others are severe because of peak overload. For example, branch 31-272 reaches 142.60% even though it appears far less frequently than 816-817 or 817-818. That difference matters for planning interpretation: persistent overloads may indicate structural congestion, while less frequent but high peaks may indicate operating conditions that require targeted mitigation.

7.3 N-0 Voltage Results

Voltage ranking was based on persistence first and severity second. In the current pre-ACCOR N-0 dataset, the voltage issue is predominantly over-voltage rather than under-voltage. This indicates that the main baseline voltage challenge in the solved cases is not widespread low-voltage weakness, but rather repeated high-voltage excursions at a set of buses that remain outside the normal planning band.

Bus	Bus name	kV	Hours violated	Dominant type	Max pu	Min pu	Average pu
10605	THICKWD	23.0	5215	High	1.2691	1.2414	1.2592
3270	SUFFIEL8	23.0	5215	High	1.2654	1.2533	1.2598
1080	WESLY C4	240.0	5215	High	1.1310	1.1228	1.1282
1655	DAWES240	240.0	5215	High	1.1239	1.1209	1.1221
1605	THICKWD	240.0	5215	High	1.1200	1.1200	1.1200
1090	MEIKLE1	138.0	5215	High	1.1176	1.1123	1.1154
204	MCMILLA4	240.0	5215	High	1.1174	1.1119	1.1143
1050	HOTCHKS7	138.0	5215	High	1.1173	1.1120	1.1152
1083	CARMON 7	138.0	5215	High	1.1168	1.1080	1.1140
1280	BRINT02	240.0	5215	High	1.1165	1.1021	1.1100

Table 5: N-0 Voltage Results

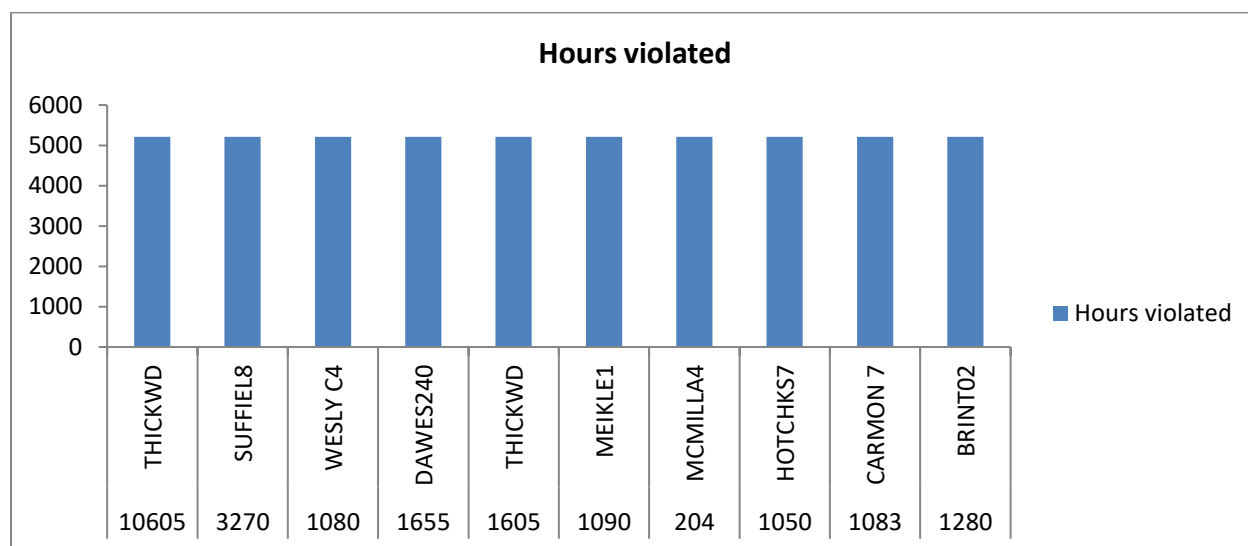


Figure 5: Top buses with persistent pre-ACCOR N-0 voltage violations.

The most severe and persistent bus in the ranked set is bus 10605 (THICKWD), with a maximum voltage of 1.2691 p.u., a minimum of 1.2414 p.u., and voltage violation in all 5215 solved hours. Several other buses - including 3270, WESLY C4, DAWES240, MEIKLE1, and MCMILLA4 - also violate the normal band in every solved hour. This consistency suggests that the present N-0 baseline contains a strong systematic over-voltage pattern rather than a small number of isolated local issues.

Because area and regional labels were only partially recoverable for the voltage results, the most defensible interpretation is bus-based rather than area-based. The report therefore prioritizes direct bus evidence over incomplete area attribution.

7.4 Critical N-0 Hours Reserved for Selected N-1 Screening

Rank	Hour	Month	Max thermal loading (%)	Thermal violations	Voltage violations	Worst violating branch	Total excess above 100 (%)
1	2025-04-11 10:00	2025-04	165.01	8	533	816-817	215.26
2	2025-04-09 09:00	2025-04	163.74	7	532	816-817	186.00
3	2025-07-26 20:00	2025-07	162.52	8	546	816-817	189.81
4	2025-01-27 06:00	2025-01	162.25	6	561	816-817	192.53
5	2025-01-09 04:00	2025-01	161.94	7	567	816-817	208.72
6	2025-12-19 03:00	2025-12	161.93	8	551	816-817	215.65
7	2025-04-25 02:00	2025-04	161.87	6	540	816-817	189.93
8	2025-05-07 23:00	2025-05	161.78	6	526	816-817	192.91
9	2025-01-09 23:00	2025-01	161.59	7	574	816-817	204.58
10	2025-09-21 20:00	2025-09	161.59	7	546	816-817	197.97

Table 6: Critical N-0 Hours for Selected N-1 Screening

The top 50 worst N-0 hours were ranked by maximum pre-ACCOR loading, then by the number of simultaneous thermal violations, and finally by total excess loading above 100%. This ranking provides a systematic basis for selected N-1 follow-up. The highest-ranked hour in the current dataset is 2025-04-11 10:00, with a maximum thermal loading of 165.01%, eight simultaneous thermal violations, and worst branch 816-817.

A notable finding is that branch 816-817 appears as the worst violating branch in all top-ranked hours shown in the table. This reinforces the conclusion that it is the dominant thermal bottleneck and a natural priority for selected N-1 contingency screening once the post-contingency results are available.

7.5 Measures of Success Against Current Evidence

Measure of success	Acceptance target	Status	Current evidence
Automated N-0 screening pipeline	8760-hour loop with structured outputs	Partially achieved	The implemented workflow produced 5215 solved hourly N-0 cases and the full report-oriented summary workbook.
Thermal violation detection	Correct identification of overloaded monitored lines	Achieved for current baseline	Thermal branch rankings, top 10 tables, and critical-hour selection were produced from the pre-ACCOR violation scan.
Voltage violation detection	Correct identification of buses outside 0.95-1.05 p.u.	Achieved for current baseline	Persistent high-voltage buses were identified and ranked from the N-0 bus violation scan.
Critical-hour screening for N-1	Selection of representative worst N-0 hours	Achieved	Top 50 worst N-0 hours were extracted and retained for selected N-1 follow-up.
ACCOR integration	Automatic congestion-relief capability with debug logging(see supplementary files	Achieved	ACCOR output evidence is now available. The gen_adj_hourly file shows non-zero generator adjustment in 7250 of 8760 hours (82.8%), with active-hour GEN-ADJ totals of 174.5-186.8 MW. The generator_outputs archive adds 217 generator-level CSV traces.

Table 7: Measures of Success

7.6 Validation Interpretation and Partial-Success Statement

The current report should be understood as a validated N-0 baseline with completed automation architecture, exported ACCOR-adjustment evidence, and a clearly defined selected N-1 follow-up pathway. It does not claim that all original deliverables are fully closed. Instead, it demonstrates substantial engineering value through a robust N-0 screening workflow, quantitative identification of

recurring thermal bottlenecks, quantitative identification of persistent voltage problems, and systematic extraction of critical N-0 hours for the next stage of study, and traceable generator-adjustment outputs from the implemented ACCOR workflow.

This partial-success framing is important for engineering credibility. A planning tool is more trustworthy when it states its limits clearly than when it over claims completeness.

8. List of Tools, Materials, Supplies, Costs, etc.

This project is a software-centered engineering study and therefore relies primarily on licensed simulation software, open-source Python libraries, and standard computing hardware. No custom hardware, laboratory fabrication, or purchased physical components were required.

Item	Purpose	Actual cost	Comment
PSS®E university license	AC power flow, ACCOR, case handling	\$0	Roadmap treated this as zero direct academic cost; industrial value is high.
Python	Automation, ETL, analysis	\$0	Matches roadmap assumption of standard installed tooling.
Pandas / NumPy	Hourly data preparation and aggregation	\$0	Open-source; used directly in the scripts.
Matplotlib / plotting assets	Figures for report and result interpretation	\$0	Open-source or generated internally.
VS Code / Spyder or equivalent IDE	Code development and debugging	\$0	Free tooling consistent with roadmap assumptions.
University lab computer / personal laptop	Batch execution and file storage	\$0	Existing resource; industrial equivalent would imply non-zero workstation cost.

Table 8: List of Tools, Materials, Supplies, Costs

9. Reflection on Ethics and Equity in our Capstone Journey

Ethically, the most important issue in this project is honest representation of what the tool can and cannot support. Because the tool is intended for planning-level screening, not operational dispatch, it would be professionally inappropriate to present its results as direct real-time operating instructions.

A second ethical issue concerns transparency in data handling. Mapping hourly AESO assets to a large PSS®E base case inevitably requires exclusions, proxy assignments, and engineering judgement. The

team's use of mapping tables, exception logs, and documented proxy rules therefore functions not only as a technical convenience but also as an ethical safeguard.

Equity enters at both the energy-system level and the team level. A neutral screening tool helps reduce bias by making bottlenecks visible before infrastructure or policy decisions are made, while the project's modular structure enabled meaningful contributions across data preparation, modelling, automation, validation, documentation, and visualization.

10. Reflection on Impact of Engineering on Society/Environment

The project's positive societal impact is that better congestion screening can improve the quality of transmission planning in a system with rapidly growing renewable penetration. When planners understand where recurring bottlenecks exist and how much curtailment they cause, they can prioritize reinforcements and operational strategies more intelligently.

The project also has clear environmental significance. Renewable curtailment represents clean energy that could have been produced but was not delivered because of network constraints. By identifying the transmission elements and operating conditions that drive that curtailment, the project makes the environmental cost of constraint visible.

The cautionary side is that simplified planning tools can be misused if their limitations are ignored. A steady-state tool does not capture dynamics, protection interactions, or all operational constraints. The correct mitigation is the one adopted in this report: explicit scope definition and transparent assumptions.

11. Reflection on Team's Approach to Project Management

The uploaded project documents show a sound management approach built around sprint-based planning, explicit milestones, and role clarity. That structure was especially appropriate for a technical project because it forced prerequisite tasks such as base-case validation, mapping, and criteria definition to be completed before computationally expensive batch execution.

Risk management was handled pragmatically. Instead of assuming that long batch simulations would run perfectly, the team explicitly identified data inconsistency, runtime burden, Python–PSS®E integration failure, and result misinterpretation as risks, and paired them with mitigation strategies such as modular scripts, structured logging, and targeted rather than exhaustive N-1 analysis.

One of the strongest management choices was the alignment between technical architecture and team structure. The modular workflow made it possible for different members to own meaningful parts of the project while still supporting integrated review.

12. Reflection Team's Approach to Economics and Cost Control

Although the project did not require direct spending, economics still influenced design decisions. The clearest example is the decision to use selected N-1 analysis rather than a full N-1 run for all hours and elements. That choice reduced compute time, shortened debugging cycles, and improved overall project feasibility.

A second example is scope control. By intentionally excluding transformer monitoring, dynamic studies, and real-time dashboard ambitions from the baseline product, the team avoided scope growth that would have diluted the value of the core deliverable.

The staged expansion path preserved in the report remains economically sound: once the baseline engine is validated further, the team can compare the incremental value of new capabilities against the effort required to implement them.

13. Reflections on Team Member's Approach to Life-long Learning

This capstone required the team to learn beyond any single course boundary. The work combines power-system modelling, transmission-planning logic, Python scripting, structured data handling, debugging discipline, and technical reporting. The project therefore demanded continuous learning rather than isolated task execution.

For individual team members, the project reinforced that modern power engineering increasingly depends on hybrid capability. Knowing AC power flow alone is no longer enough, and knowing Python alone is not enough either. Engineers must connect domain judgment with data and automation.

This learning approach is valuable beyond the capstone itself. Future work in utilities, consulting, planning, or research will likely involve new tools, new datasets, and evolving modelling standards. The habits of documenting assumptions, validating automation, and communicating limits are therefore long-term professional assets.

14. Appendix

Appendix A

Load Hourly Mapping (Implementation Summary)

Hourly loads are modelled by anchoring to the solved 2025 PSS®E base case and applying historical AESO hourly demand variability. Base-case constant-power loads (MW/Mvar) are aggregated by PSS®E Area to form Area totals. AESO 2024 hourly load data are converted to normalized hourly shapes by Area, and the study-year demand level (2026) is represented by scaling the base-case Area totals using an annual growth proxy of 1.9%. Hourly Area scaling factors are then applied to base-case bus loads within each Area, preserving the base-case power factor via constant-PQ scaling.

Key assumptions:

- Fixed network snapshot: network topology and device settings remain as in the solved 2025 base case; only loads vary hourly.
- Spatial allocation preserved: within each Area, bus-level load shares remain proportional to the base case (no intra-area load relocation).
- Constant PQ / constant power factor: reactive power scales proportionally with real power to preserve base-case power factor.
- Shape representativeness: 2024 hourly Area demand profiles are assumed representative for shaping 2026 hourly variability.
- Uniform growth proxy: a uniform annual growth factor is applied from 2025 to 2026.
- 8,760-hour alignment: the time series is treated as a non-leap-year (8,760 hours).

Load scaling equations:

A.1 Base-case Area totals (from PSSE base case)

For each Area a , base-case totals are computed by summing all load records i assigned to that Area:

$$P_0(a) = \sum_{i \in a} P_{i,0}, Q_0(a) = \sum_{i \in a} Q_{i,0}$$

A.2 Normalized hourly shape (from AESO historical data)

Using AESO hourly Area demand $P_{2024}^{AESO}(a, t)$, a normalized shape is formed:

$$P_{2024}^{AESO}(a) = \frac{1}{8760} \sum_{t=1}^{8760} P_{2024}^{AESO}(a, t)$$

$$s(a, t) = \frac{P_{2024}^{AESO}(a, t)}{P_{2024}^{AESO}(a)}$$

This isolates hourly variability from absolute demand level.

A.3 Growth multiplier (High Electrification proxy)

With annual growth rate $g = 0.019$, the multiplier from 2025→2026 is:

$$G_{2026 \leftarrow 2025} = (1 + g) = 1.019$$

A.4 Hourly Area scaling factor

The hourly scaling factor applied to the base case is:

$$sf(a, t) = G_{2026 \leftarrow 2025} \cdot s(a, t)$$

A.5 Bus-level hourly loads (proportional allocation)

For each base-case load record i in Area a :

$$P_i(t) = P_{i,0} \cdot sf(a, t)$$

$$Q_i(t) = Q_{i,0} \cdot sf(a, t)$$

This preserves the base-case spatial allocation and power factor while matching the AESO-derived hourly Area variability and the study-year growth level.

Appendix B

Generator Hourly Mapping (Implementation Summary)

Hourly generation is modelled by anchoring to the solved AESO 2025SP PSS®E base case and applying historical AESO hourly metered generation profiles to mapped generators. Each hourly asset is mapped (where possible) to one or more base-case PSS®E machines using a Name-Match table (asset name plus location/area cues). Mapped hourly MW setpoints are applied subject to modelled capability limits and in-service status. Hourly assets not represented in the 2025SP machine list are excluded and recorded in an exception log (e.g., BPW1, ALP1). Conversely, generators present in the base case but lacking hourly profiles are assigned proxy dispatch shapes from similar technology in the same (or nearest) planning area and scaled to modelled capability.

Key assumptions:

- Fixed network snapshot: network topology and device settings remain as in the solved 2025SP base case; only generator dispatch varies hourly.
- Case-consistent resource set: assets not modelled in the 2025SP base case are not injected (excluded and logged) to avoid non-transparent balancing through slack/tie responses.
- Real-power driven: hourly profiles set MW dispatch; reactive/voltage control remains as modelled in the base case (no retuning).
- Capability enforcement: applied dispatch is limited by modelled capability limits and unit in-service status.
- Proxy representativeness: for modelled units without hourly data, proxy shapes are assumed representative within the same technology class and planning area.
- 8,760-hour alignment: the time series is treated as a non-leap-year (8,760 hours).

Generator dispatch equations:

B.1 Base-case generator capability (from PSS®E base case)

For each mapped PSS®E machine i , extract:

- P_{MAX_i} , P_{MIN_i} (if used), in-service status, bus/area identifiers.

B.2 Direct hourly dispatch (hourly asset mapped to one machine)

For an hourly asset g mapped to machine i :

$$P_i(t) = \begin{cases} \min(P_g(t), PMA X_i), & \text{if status}_i = 1 \\ 0, & \text{if status}_i = 0 \end{cases}$$

B.3 Multi-machine split (one hourly asset mapped to multiple machines at same POI)

If asset g maps to machines $i \in \mathcal{M}$ representing the same plant/POI, allocate by capability weights:

$$w_i = \frac{PMA X_i}{\sum_{k \in \mathcal{M}} PMA X_k}$$

$$P_i(t) = \min(P_g(t) \cdot w_i, PMA X_i)$$

(If clipping occurs on any machine, redistribute remaining MW to other machines with headroom.)

B.4 Proxy dispatch (modeled in PSS®E but no hourly series)

Select a proxy shape $s(t) \in [0,1]$ from a similar generator (same technology; same/nearest area). Scale to modeled capability:

$$P_i(t) = \min(\alpha_i \cdot s(t), PMA X_i)$$

where α_i is chosen to match either (i) a target average output level, or (ii) the base-case $PGEN_i$ (flat/near-flat conventional dispatch).

15. References

1. AESO, Hourly Load by Area and Region (market and system reporting data request), accessed February 2026.
<https://www.aeso.ca/market/market-and-system-reporting/data-requests/hourly-load-by-area-and-region/>
2. AESO, Generation Metered Volumes (hourly generator metered volumes dataset), accessed February 2026.
<https://www.aeso.ca/market/market-and-system-reporting/data-requests/historical-generation-data/>
3. AESO, Base Cases (planning snapshots used for studies), accessed February 2026.
4. AESO, Long-term Outlook (High Electrification scenario; energy consumption CAGR proxy of 1.9% per year), accessed February 2026.
<https://www.aeso.ca/grid/grid-planning/forecasting/2024-long-term-outlook/>